

Stackcraft: Adapting Clef-flash to a Falling-Block Decision Task

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Abstract

Can a general decision model learn a useful game policy from a small synthetic dataset on one workstation? Stackcraft studies this question in a deterministic, turn-based falling-block game. A Clef-flash model was adapted using rank-4 LoRA and its full native decision head, with 827 training positions and 215 validation positions labeled by a bounded search teacher. One training run produced two checkpoint candidates; validation negative log likelihood selected the second. Across 200 previously reserved piece sequences per policy, the selected model cleared 16.81 lines per game, versus 0.07 for unchanged Clef-flash. The paired improvement was 16.74 lines, with a 95% bootstrap interval of [15.74, 17.77]. However, a fixed arithmetic heuristic cleared 76.53 lines and was substantially faster. All 1,000 evaluated episodes completed without inference errors or invalid decisions. The study demonstrates task adaptation under a constrained local budget, while showing why improvement over a weak base model is insufficient evidence of practical superiority. Code, data, learned parameters, recorded games, and verification evidence are public. This is a technical report, not a peer-reviewed publication.

1. Motivation and scope

A falling-block game offers a concrete test of decision quality: choices alter the board, mistakes accumulate, and complete games expose failures that isolated classification scores can hide. Stackcraft asks whether a small imitation-learning dataset can improve a general decision model, and whether that improvement is competitive with a simple task-specific policy. The project also makes the experiment inspectable through replayable games and milestone tutorials.

Clef-flash is a roughly nine-billion-parameter model derived from Qwen3.5-9B. Its native interface accepts a state and typed questions, then produces probabilities over supplied options through a joint decision head. This avoids requiring a generated command or a text parser to select a legal move. Stackcraft uses text and structured state only; it does not test the model's visual capabilities. The implementation follows the pinned upstream interface. [1]

The contribution is an engineering case study with released evidence. No new adaptation algorithm, general game-playing advance, or superiority over specialized game agents is claimed. The central distinction is between learning to play better than the unchanged model and building the best bot for these rules.

2. Game and policy interface

The board has ten columns and twenty rows. Seven tetromino shapes arrive in seeded seven-bag sequences: each shuffled bag contains every shape once. A turn chooses a legal orientation and column, then drops the piece vertically. The rules omit hold, wall kicks, tucks, T-spin bonuses, and real-time movement requirements. Clearing one, two, three, or four lines awards 100, 300, 500, or 800 points. Lines cleared per episode is the primary outcome.

Every policy receives the board, current piece, one next-piece preview, and the same legal placement options. Hidden future pieces, the episode seed, sequence index, and generator state are excluded from the observation. Python owns the authoritative transitions, so browser play, offline tournaments, and replay validation use the same rules. This is a structured placement task rather than a benchmark of full competitive Tetris.

Five policies were evaluated. **Native base** retains unchanged Clef-flash weights and its original BF16 head. **FP32 base** uses unchanged weights with the FP32 head wrapper used for training. **Trained** combines the selected LoRA adapter with the learned FP32 head. All three share the pinned backbone, observation encoding, option ordering, BF16 backbone precision, and 4,096-token limit. Full observations are checked against that limit before inference.

Random samples uniformly from legal options using a recorded random generator. **Heuristic** greedily evaluates the resulting board with a fixed formula: negative aggregate column height, minus four times the number of holes, minus adjacent-column height differences, plus eight times the number of cleared lines. A hole is an empty cell below an occupied cell in its column. The heuristic does not use the available preview; its weights were frozen before final testing. It is distinct from the lookahead teacher that generated training labels. [3]

3. Synthetic data and adaptation

Data construction

The dataset contains 827 training positions from seeds 10000–10023 and 215 validation positions from seeds 20000–20005. Collection cycles among random, greedy-heuristic, and search-expert behavior, recording at most forty moves per episode. This exposes the learner to boards produced by different policies rather than only successful expert trajectories. Every episode belongs to one split. Occupancy-normalized duplicate detection removed six repeated training positions and two validation positions that overlapped the training split.

For each retained observation, the teacher enumerates legal placements of the current piece and the one visible preview, then evaluates line clears and board quality. It cannot inspect unseen pieces. Its chosen action is a deterministic, bounded-search recommendation, not a globally optimal label. The released dataset records legal choices, teacher values, labels, source identity, and split provenance. It is programmatically generated synthetic data; it is not a corpus of human gameplay or human-certified optimal decisions. The reserved test seeds 30000–30199 were excluded from collection and checkpoint selection. [4]

Training configuration

LoRA adds trainable low-rank weight updates while retaining frozen backbone parameters. [2] Here it was applied to the text backbone alongside full training of Clef's native joint head. A single seed-42 run used rank 4, alpha 8, zero adapter dropout, a BF16 unquantized backbone, and an FP32 head. The combined trainable parameter count was 132,582,404. This count includes the full head; “LoRA” should not be read as implying that only a tiny adapter was trained.

Training ran for exactly two epochs. The objective combined cross-entropy with 0.05 label smoothing and a Brier-loss term weighted by 0.1. AdamW used a learning rate of 0.00001 and weight decay 0.01. Batch size was one, gradient accumulation was eight, and gradient norm clipping was 1.0. Each epoch processed all 827 positions in 104 optimizer steps. The final accumulation group used its actual three examples for normalization. Recorded hashes confirmed that frozen parameters remained unchanged while intended trainable parameters changed.

Selection and reload checks

The two epoch checkpoints were the entire preregistered selection search. Eligibility required complete, finite predictions on all 215 validation positions; selection minimized mean target negative log likelihood (NLL), with exact ties going to the earlier epoch. Both candidates qualified, and epoch 02 was selected. Teacher agreement and Brier scores were diagnostics, not alternative selection criteria. The local preregistration was preserved in source; it was not a signed external registration.

Condition	Teacher agreement	Mean NLL	Mean Brier
Native base	21/215 (9.77%)	2.952702	0.923173
FP32 base	22/215 (10.23%)	2.952393	0.923102
Epoch 01	89/215 (41.40%)	1.860870	0.722190
Epoch 02 (selected)	92/215 (42.79%)	1.775923	0.708980

Table 1. Validation results on all 215 positions. Lower NLL and Brier values are better. The second epoch was selected before test games were opened.

The selected model agreed with 92 teacher actions, or 42.79%. That percentage is not a game-quality score. Alternative moves may be useful, teacher tie-breaking can affect agreement, and a wrong move changes the states encountered later. Complete held-out games therefore provide the behavioral evaluation.

The checkpoint saves both the adapter and native head. A fresh-process reload matched four fixed training-position reference distributions exactly, with a maximum absolute probability difference of 0.0 against a declared tolerance of 0.0001. The same four-reference check later passed using anonymously downloaded published code and weights. These checks establish serialization consistency on those references, not correctness on every board. [3], [5]

4. Held-out evaluation

Each policy played the same 200 held-out sequences, seeds 30000–30199, with a cap of 200 placed pieces per episode. The tournament therefore contains 1,000 episodes. A cap hit means survival for at least 200 pieces; it is not a loss on piece 200. No model, prompt, policy, dataset, or checkpoint was tuned after opening this final test.

Comparisons subtract outcomes within each seed. The analysis resamples the 200 paired episode differences with replacement 10,000 times, using bootstrap seed 2026, and takes percentile 95% intervals. Resampling episodes retains dependence among decisions in the same game. The primary contrast is trained minus native base in lines cleared. Score and survival intervals are descriptive secondary results without a multiplicity correction.

Player	Mean lines	Median lines	Mean score	Mean pieces	Cap hits
Native base	0.070	0	7.0	26.020	0/200
FP32 base	0.105	0	10.5	26.000	0/200
Trained	16.810	16	1745.5	83.565	0/200
Random	0.140	0	14.0	26.645	0/200
Heuristic	76.530	77	8306.5	200.000	200/200

Table 2. Held-out outcomes for 200 episodes per policy. Survival is pieces placed. All five policies recorded zero inference errors and invalid decisions.

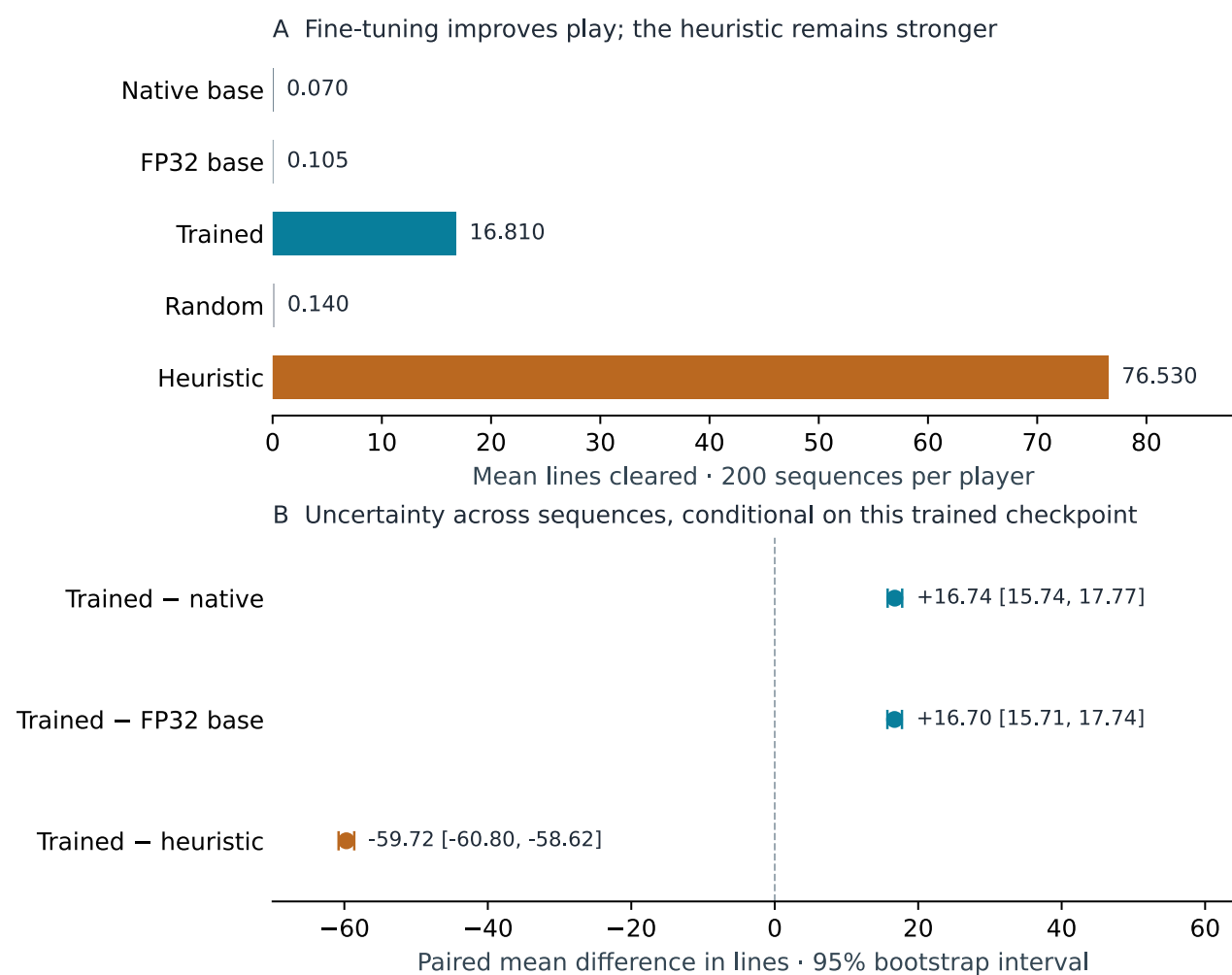


Figure 1. Mean lines cleared and paired differences for the selected checkpoint. Intervals describe variability across held-out sequences for this checkpoint, not variation across independently trained models.

The trained model cleared 16.81 lines on average, compared with 0.07 for native base: a gain of 16.74 lines, 95% interval [15.74, 17.77]. The gain against FP32 base was 16.705 [15.705, 17.740]. The FP32-minus-native difference was only 0.035 [−0.020, 0.090], so changing head precision alone does not explain the observed adaptation gain. This comparison does not separate learned adapter effects from learned head effects; both were optimized together.

The heuristic remained substantially stronger, clearing 76.53 lines on average. The trained-minus-heuristic difference was −59.720 [−60.805, −58.619875]. Every heuristic episode reached the cap; no other policy did. Its unrestricted survival is consequently unknown. No

failed episodes were removed. The frozen analysis would assign zero lines, score, and survival to an errored episode while retaining its partial outcome separately; because there were no errors, adjusted and observed results coincide.

An independently implemented audit reconstructed all 1,000 saved episodes, checked observations and legal choices, verified source and checkpoint bindings, and recomputed all twelve paired intervals across lines, score, and survival. It matched the published results. This was a separate code-path check by another AI agent, not external scientific replication. [3]

5. Runtime and practical value

Experiments ran on an NVIDIA RTX 5090 with roughly 32 GB of VRAM, an AMD Ryzen 9 9950X3D, and about 123 GiB of host RAM. The recorded environment used Python 3.13.16, Torch 2.14.1+cu130, CUDA 13.0, Transformers 5.18.0, and PEFT 0.21.2. Torch used eight CPU threads and TF32 was disabled.

Player	Mean ms	Median ms	p95 ms	Decisions
Native base	149.109957	139.381274	244.916266	5,204
FP32 base	150.188366	140.968013	247.879591	5,200
Trained	206.814401	171.844287	309.000088	16,713
Random	0.001497	0.001250	0.002879	5,329
Heuristic	0.198510	0.139489	0.280528	40,000

Table 3. Per-decision timings over each policy's actual tournament workload. Initialization is excluded; no warmup calls within the tournament were removed.

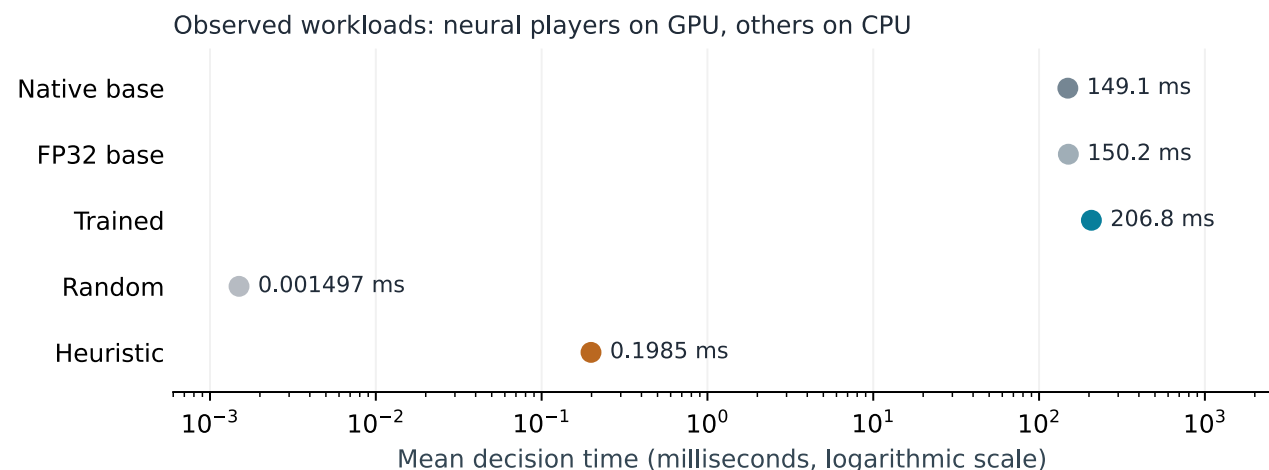


Figure 2. Observed mean decision latency. Neural policies use the GPU; random and heuristic policies use the CPU. Different policies visit different boards, so this is not a controlled same-state speed benchmark.

Decision timings include encoding and probability transfer, but exclude model loading, rendering, and replay serialization. The trained mean was 206.814 ms, versus approximately 0.198510 ms for the heuristic: about 1,042 times longer on these workloads. The trained policy also saw longer inputs on average than either unchanged model. These measurements support choosing the heuristic for a practical bot under the tested rules; they do not isolate a hardware-independent inference speed ratio.

Training took 1,511.25 seconds, or 25.2 minutes, including checkpoint and hash work. Peak CUDA allocation was 23.23 GB (21.64 GiB), with 24.18 GB reserved. The final evaluation wrapper took 5,103.11 seconds, or 85.1 minutes, including loading, reporting, and service restoration checks. These durations exclude previous development, validation, and data generation. Energy use and monetary cost were not measured. [3]

6. Limitations and next experiments

The strongest limitation is the single training seed. The bootstrap intervals measure variability across episode sequences for one selected model; they do not quantify training instability. The dataset is small and synthetic, and its teacher optimizes a short-horizon approximation. Collection stops after at most forty moves, whereas test games can last two hundred. Errors may therefore lead the policy into poorly represented states, although this study does not establish a causal failure mechanism.

Only one structured encoding, one adaptation configuration, and two epoch candidates were studied. The weak unchanged-model result characterizes that pinned model under this interface, not every possible use of Clef-flash. The rules omit many skills relevant to real-time Tetris. The heuristic's universal cap hits limit claims about its full survival distribution. None of these findings establishes performance on a different game, board representation, or deployment platform.

Useful follow-up studies could compare head-only training with joint head/LoRA training, repeat training across seeds, and evaluate a compact game-specific student against the same inexpensive heuristic. Gathering recovery states from the learned policy could test whether broader state coverage improves complete games. Those are proposed experiments, not measured improvements. Any follow-up should declare its selection procedure and reserve fresh test sequences rather than repeatedly optimizing against the published test set.

7. Reproducibility and availability

The public code repository includes the game, experiment scripts, tests, a locked uv environment, and tutorials. The model release includes the adapter, learned head, reproduction source, and a lossless evidence archive with 1,017 records. The archive's inventory preserves uncompressed file hashes. Anonymous downloads of the pinned model and dataset passed byte and schema verification; all archive entries were read and hashed. The published reload check used downloaded source and checkpoint files. [5]

The frozen model release is c4272310bd6c63ee97a941255abf8f36ff162229; the dataset release is 211708dd56f1a8af711c060c2ab1ecab35e7166d. The bundled release source is 82a8854258f228a214f0964bbbd3d2b52e2be1e5. These identify the experimental artifacts, not the later additive PDF publication. The upstream Clef-flash revision is 17f0b0ad64efb65d273590632833508766b2aae6. [1], [4], [6]

Provenance distinguishes stages rather than assigning every result to the release commit. Data generation used commit 70d84bd8dc5d4a60f3b96455a57d9e6f9416d109. Training recorded d9fcd05c02e9327d4d791f2338725092ced8bdd with a dirty working tree and explicit source-file hashes; that commit alone is not the full training source identity. Final evaluation used clean commit e524f0c15c2dd60ebfcd1374d66836c79e1e6dd6. The retained reports and source hashes are the authoritative record of these distinctions. [3]

The local browser demo pairs live human play with clearly labeled recordings from the actual base, trained, and heuristic policies. Playback speed is not live neural inference speed. Hosted CPU deployment is deferred; no public running Space is claimed. Readers can inspect evidence and replay games without retraining or renting a GPU.

Assistance disclosure

AI coding agents assisted with implementation, experiment orchestration, documentation, manuscript preparation, and computational review under the project owner's direction. The separate audit used another agent and an independent implementation, but was part of this project. Synthetic labels came from the programmatic search teacher. No external peer review, independent laboratory replication, or exhaustive human labeling review is claimed.

References

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